Sagar Mitra — Complete Video Demonstration Script

Smart India Hackathon (SIH 2026) | Problem Statement: SIH26176

Organization: Indian Space Research Organisation (ISRO) · ORCA Marine EcoSystem Reasoning with Collaborative Agents

Scene 1 · The Core Challenge & The ISRO Mission (SIH26176)

■ **[SHOW ON SCREEN]**

Start on login.html in dark mode, showing the ocean video player on the right and login card on the left.

■ **[SAY - SPOKEN VOICE]**

India has over seven thousand five hundred kilometres of coastline supporting millions of livelihoods. Yet today, ocean science remains fragmented. A small-scale fisherman heads out at dawn relying purely on guesswork about where fish might be or whether a squall is brewing. At the same time, marine scientists are stuck wrestling with raw satellite datasets, and school students have no engaging way to learn about the sea.

Our human motivation was simple: *why should world-class satellite data remain locked in technical formats when everyday citizens need it most?* That is why we built **Sagar Mitra** for the Smart India Hackathon under Problem Statement **SIH26176** by the **Indian Space Research Organisation (ISRO)**: *'ORCA Marine EcoSystem Reasoning with Collaborative Agents'*. Our mission is to translate complex Earth Observation data into intuitive, human-centric intelligence for every ocean stakeholder.

Scene 2 · User Onboarding: Personas & Domain Curation

■ **[SHOW ON SCREEN]**

Click 'New here? Create account'. Briefly pass credentials to Step 2 ('Select your persona') showing the 7 roles (Primary Student, High School, Researcher, Fisherman, Coastal Resident, Authority, Others). Then proceed to Step 3 ('Topics you want to read') showcasing the 8 domain selection cards.

■ **[SAY - SPOKEN VOICE]**

We realized that a one-size-fits-all dashboard fails because a researcher analyzing thermal anomalies has completely different priorities than a fisherman battling strong swells. So right in our onboarding flow, you select from seven distinct personas: Primary Students, High Schoolers, Researchers, Fishermen, Coastal Residents, Port Authorities, or General Explorers.

Immediately after, in Step Three, you choose your ocean domains — such as Pelagic Fisheries, Remote Sensing, ARGO buoys, or Cyclone Meteorology. The human thought here is personalization from day one: rather than cluttering your screen with irrelevant data, Sagar Mitra tailors its interface, language complexity, and background telemetry specifically to your daily reality. Let us click 'Complete & Dive In' to enter the main deck.

Scene 3 · Homepage Dashboard & Persistent Action Controls

■ **[SHOW ON SCREEN]**

Landing on main.html. Hover over top-right buttons in order: Navigation, Morning Dispatch, News Feed, Switch Role, Theme Toggle, and Logout. Point out the rotating multilingual title.

■ [SAY - SPOKEN VOICE]

Welcome to the homepage. Notice how the hero title rotates 'Sagar Mitra' across ten Indian languages — including Hindi, Marathi, Bengali, Tamil, Telugu, and Gujarati. Our team actually went on the ground to interview a local fisherman on the Konkan coast. We learned that the biggest barrier to technology adoption is language. If an alert is in complex English, it gets ignored.

Across the top right, we created a universal control deck designed for quick, distraction-free access: • This **wave icon** opens our draggable navigation bar so you can move your menu wherever it does not obstruct your data. • This **orange document icon** is the Morning Dispatch — built because harbor operators and fishermen need a quick, printable PDF summary of wave swells and fishing zones before casting off, without digging through menus. • This **yellow newspaper icon** opens an ocean News Feed pulling live scientific and maritime updates matching your chosen domains. • And our **green persona switcher** lets you switch roles on the fly without re-logging, accompanied by our **dark/light theme toggle** for night-time bridge navigation and our **logout button**.

Scene 4 · About Sagar Mitra: Ground Research & Persona Dynamic Slide

■ [SHOW ON SCREEN]

Navigate to about.html. Show field observation documentary video on Slide 1. Then on Slide 2, switch persona to 'Researcher' and watch content re-render dynamically.

■ [SAY - SPOKEN VOICE]

Here on the About page, Slide One shares our on-the-ground documentary from the docks. Slide Two embodies our adaptive philosophy: if you are logged in as a Fisherman, it provides direct advisories on nearshore wave heights, safe harbor depths, and potential fishing zones. If you switch to Researcher, the exact same section automatically adapts to explain multi-agent geospatial reasoning, satellite arrays, and thermal anomaly modeling. The user never feels out of place.

Scene 5 · Ocean Intelligence Map: Safety Dock & ML Overlays

■ [SHOW ON SCREEN]

Navigate to map.html. Show full-screen map. Point out top coordinates HUD. On left Safety Dock, demonstrate Anchor Drift, Hazard Report modal, SOS button, and EEZ boundary check. On right, click PFZ, ARGO, and Route Planner.

■ [SAY - SPOKEN VOICE]

Now, let us explore the Ocean Intelligence Map. The top HUD gives captains and scientists live coordinates and zoom depth.

On the left, we built a dedicated **Safety Dock** based directly on fishermen's real safety concerns: • The **Anchor button** activates an Anchor Drift Sentry — because boats drifting at night while crews sleep cause severe accidents; our system alerts them the second they drift past fifty metres. • The **hazard triangle** lets any vessel report floating ghost nets or oil slicks into a crowdsourced mesh, protecting fellow sailors from fouled propellers. • The **red SOS button** logs an emergency distress ping with exact GPS coordinates to our incident database. • And the **geofence shield** checks whether a boat is within Indian Exclusive Economic Zone territorial waters, preventing accidental international border crossings and keeping boats out of protected marine sanctuaries.

On the right, we have eleven scientific layers: • Clicking **PFZ** activates our machine learning Potential Fishing Zone model. Instead of giving a black-box suggestion, we provide Explainable AI feature attributions so fishermen can trust *why* a zone was recommended based on temperature and chlorophyll. • Clicking **ARGO** visualizes over four thousand live profiling floats, allowing oceanographers to inspect temperature and salinity soundings down to two thousand metres with one click. • And clicking **Route Planner** launches our multi-objective **Reinforcement Learning** engine. Rather than simply drawing a straight line, it balances fuel savings in Eco-Saver mode with catch probability in Harvest mode, accompanied by an Explainable AI nautical logbook explaining every waypoint decision.

Scene 6 · Chatbot: Split Screen & 4 Live Persona Demonstrations

■ [SHOW ON SCREEN]

Navigate to chatbot.html. Show workspaces in sidebar. Click 'Split Map' at top right to reveal 3/5 chat and 2/5 interactive map viewport. Then run the 4 live test prompts.

■ [SAY - SPOKEN VOICE]

Finally, we enter the core intelligence engine — the Sagar Mitra Chatbot. Conversations are organized into Research Workspaces in the sidebar so scientists never lose field notes. Clicking **Split Map** at the top right splits your screen into a dual-viewport: conversational reasoning on the left and an interactive spatial map with climate trend charts on the right. Let us test how our collaborative agents respond across four real-world scenarios:

■ DEMO 1 · Fisherman Persona (Native Language Query)

Type in chat: "Kya aaj Mumbai ke paas machhali pakadna safe hai?"

[Say while sending query]: 'We type this query in Hindi asking if it is safe to fish off Mumbai today. Our human intent here is safety and dignity — answering directly in Hindi with wave heights, wind warnings, and nearby fishing coordinates. By delivering life-saving ocean intelligence in their mother tongue, we ensure safety advisories are actually trusted and followed.'

■ DEMO 2 · Primary Student Persona (Simplified Analogies & Text-to-Speech)

Type in chat: "What is a wave? Explain like I am 8 years old."

[Say while switching role & sending]: 'Next, switching to Primary Student, we ask: what is a wave? The thought here is building curiosity, not intimidation. Our AI drops textbook jargon and uses a fun, relatable analogy about shaking a jump rope. And with one tap on the speaker icon, text-to-speech reads the explanation aloud — bridging educational gaps for kids in coastal schools where science teachers are scarce.'

■ DEMO 3 · Researcher Persona (AI Satellite Visual Generation)

Type in chat: "Generate a satellite image of a chlorophyll bloom in the Arabian Sea during monsoon."

[Say while switching role & sending]: 'Now, switching to Researcher persona, we request a satellite visual of a monsoon chlorophyll bloom. Scientists often struggle to quickly visualize complex ocean phenomena for presentations or publications. Our text-to-image AI model generates an oceanographic satellite visual directly within the conversation thread in seconds — eliminating the need for graphic design tools.'

■ DEMO 4 · Researcher Persona (Collaborative Multi-Agent Pipeline & XAI)

Type in chat: "What is the SST anomaly near Lakshadweep right now and how does it affect tuna distribution?"

[Say while sending query]: 'Lastly, we submit a complex oceanographic question. Watch our collaborative multi-agent pipeline flowchart light up live! Instead of one giant AI guessing, specialized agents divide and conquer — the planner schedules tasks, the thermal agent inspects temperatures, the biological agent tracks chlorophyll, and the synthesizer unifies the findings. We get back precise sea surface temperatures, tuna distribution projections, and an Explainable AI attribution chart showing the exact weighting behind the result. This is true collaborative agent reasoning in action.'

Scene 7 · Composer Controls & Closing Remarks

■ [SHOW ON SCREEN]

Pan across composer bar tools: paperclip (PDF research attachment), sensors, reasoning depth selector (Surface/Deep/Abyss), and voice input microphone. Conclude on the Sagar Mitra branding.

■ [SAY - SPOKEN VOICE]

In the composer bar, researchers can attach scientific PDFs for context retrieval, tap live ocean sensors, tune reasoning depth from quick surface scans to deep analytical deductions, or speak queries via voice.

Every single feature in Sagar Mitra was engineered with a clear human purpose: saving fuel and lives at sea for our fishermen, accelerating research for scientists, and inspiring the next generation of ocean stewards. When satellite Earth Observation data meets collaborative AI, we unlock a safer, smarter ocean for India.

Thank you!